

LECTURE 10: REGRESSION MODELS

Key concepts:

- Conjugate prior and posterior for normal regression
- Conditional inference
- Prior elicitation
- Model selection

1. INTRODUCTION

Most of the results for linear regressions are natural extensions of the results for means of normal distributions. These notes summarize the main theoretical results (without proofs). We will concentrate on applications and model selection.

2. CLASSICAL REGRESSION

The general linear model is

$$y = X\beta + \varepsilon \quad \varepsilon \sim \text{Normal}(0, \sigma^2)$$

where X is a fixed (non-stochastic) matrix of order $n \times K$ of rank K . This is equivalent to assuming that

$$y \sim \text{Normal}(X\beta, \sigma^2 I_n).$$

The MLE of β is

$$\hat{\beta} = (X^T X)^{-1} X^T y.$$

The MLE of σ^2 is

$$\hat{\sigma}^2 = (y - X\hat{\beta})^T (y - X\hat{\beta}) / n,$$

which is biased. An unbiased estimator of σ^2 is

$$s^2 = \frac{(y - X\hat{\beta})^T (y - X\hat{\beta})}{n - K}.$$

If X has full column rank, then

$$\hat{\beta} \sim \text{Normal}(\beta, \sigma^2 (X^T X)^{-1}) \quad \frac{(n - K)s^2}{\sigma^2} \sim \chi^2(n - K)$$

and $\hat{\beta}$ and s^2 are independent.

A linear hypothesis $H_0 : c^T \beta = c_0$ can be tested using the test statistic

$$t = \frac{c^T \hat{\beta} - c_0}{s \sqrt{c^T (X^T X)^{-1} c}}$$

Under H_0 , t has a Student t distribution with $n - k$ degrees of freedom. Large absolute values lead to rejection of H_0 .

3. BAYESIAN REGRESSION

The results for regression closely follow the corresponding results for means of a normal distribution.

We consider the diffuse prior $(\beta, \sigma^2) \propto 1/\sigma^2$. The posterior distribution of (β, σ^2) factors into

$$\sigma^2|y \sim \text{Scaled-Inverse } \chi^2(n-k, s^2) \quad \beta|\sigma^2, y \sim \text{Normal}(\hat{\beta}, \sigma^2(X^T X)^{-1})$$

To sample from this distribution, perform the following steps:

1. Draw $u \sim \chi^2_{n-k}$ and set $\sigma^2 = (n-k)s^2/u$.
2. Draw $\beta \sim \text{Normal}(\hat{\beta}, \sigma^2(X^T X)^{-1})$.

It is generally a bad idea to compute $(X^T X)^{-1}$. Better to factor $X = QR$ where Q is orthonormal and R is upper triangular (in R, use `qr(X)`), compute $\hat{\beta} = R^{-1}Q^T y$, draw $z = (z_1, \dots, z_K)^T \sim \text{Normal}(0, I_K)$ and set $\beta = \hat{\beta} + \sigma R^{-1}z$.

4. CONDITIONAL INFERENCE

The classical regression model assumes fixed X , as in ANOVA when the columns of X are indicators for the groups (subpopulations) from which the observations were drawn, with sample size fixed by design. However, in experimental studies, the elements of X are usually treatment effects (determined by randomization). In observational studies, the process determining the covariates is unknown and must be modeled. That is, there is a probability distribution for X , say $p(X|\phi)$, where ϕ are the parameters of this process, and then the model $y|X \sim \text{Normal}(X\beta, \sigma^2 I_n)$ is now assumed to hold conditional upon X .

Generally, unless the parameters of X overlap with the parameters of the y process, (β, σ^2) , it makes sense to conduct inferences conditional upon X . (If the parameters do overlap, then there is a loss of efficiency from modeling y conditional upon X compared to modeling the joint distribution of (X, y)). An example illustrates the problem.

Suppose (x, y) are i.i.d. with $x \sim \text{Bernoulli}(\phi)$ and $y|x \sim \text{Normal}(\beta x, 1-x)$. Conditional on x , either $y = \beta$ (if $x = 1$) or $y \sim \text{Normal}(0, 1)$ and is uninformative about β (if $x = 0$). In the former case, we *know* β , while in the latter the observation is uninformative. This is what the standard Bayesian analysis conditional upon x and y would give us.

The situation is more complicated for frequentist inference. If we condition on x , the classical and Bayesian inferences are similar. However, if we do not condition upon x , but ϕ is known, then $\mathbb{E}(y) = \beta\phi$, so y/ϕ is an unbiased estimator of β . Not a very good estimator, but unbiased nonetheless.

5. PRIOR ELICITATION

In most cases, it takes some work to convert vague prior information into a suitable prior. For this reason, we prefer not to exaggerate our knowledge and use weakly informative priors.

However, in some regression analyses, the covariates are highly correlated and we may want to “solve” the collinearity problem by introducing some prior information. The usual way to do this is

to determine a plausible range for regression coefficients and treat these as 95 percent prior credible intervals. If we divide the range by four, we obtain an approximate prior standard deviation.

It is desirable to deviate the regressors from their (sample) means, so the intercept becomes (approximately) the mean of y . We can then use a flat improper prior for the intercept.

Priors for variance parameters are more challenging, but usually less consequential for standard (non-hierarchical) models.

6. MODEL SELECTION

In the regression context, “model selection” usually refers to the choice of which regressors to include in the model. One Bayesian approach to model selection is to write the model as

$$y_i = \beta_0 + \beta_1 z_1 x_{i1} + \cdots + \beta_K z_K x_{iK} + \varepsilon_i$$

where $z_1, \dots, z_K$ are unknown “parameters” determining which variables are to be included. We can then assign a prior to $z_1, \dots, z_K$, *e.g.*

$$P(z_1, \dots, z_K) = \prod_{k=1}^K p_k^{z_k}$$

where p_k is the prior probability that x_{ik} belongs in the model (and we have assumed independence). There is quite a bit of analytical work on the posterior odds with Zellner’s g -prior, but the posterior can be simulated easily using MCMC.